

SMART WINDOW AUTOMATION SYSTEM

A Mini Project Report

Submitted to the APJ Abdul Kalam Technological University

in partial fulfillment of requirements for the award of degree

Bachelor of Technology

in

Electronics and Communication Engineering

by

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CERTIFICATE

This is to certify that the report entitled **Smart Window Automation System** submitted by **Adithya M S** (NSS22EC014), **Alan Sunny** (NSS22EC025), **Anshif K** (NSS22EC036) & **Krishnadas T R** (NSS21EC044) to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Electronics and Communication Engineering is a bonafide record of the miniproject work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

We hereby declare that the Miniproject report **Smart Window Automation System**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Dr.Lija Arun, Assistant Professor, Department of Electronics and Communication Engineering, NSS College of Engineering, Palakkad.

This submission represents our ideas in our words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Palakkad
01-04-2025

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Abstract

The increasing demand for intelligent home automation has led to the development of cost-effective and adaptive systems that enhance both comfort and safety. This paper presents an advanced Smart Window System powered by Arduino Uno, designed to autonomously regulate ventilation and environmental adaptation. By integrating multiple sensors, the system dynamically responds to various environmental conditions, ensuring energy efficiency and user convenience.

The Smart Window System operates based on real-time data from sensors detecting rain, smoke, gas leakage, air conditioning (AC) status, and ambient light levels. When gas leakage or smoke is detected indoors, the window automatically opens, allowing ventilation to mitigate potential hazards. Conversely, when rain is detected or the AC is activated, the window closes to prevent water entry and maintain cooling efficiency, optimizing energy consumption. Additionally, in low-light conditions, an automated mosquito net deploys, offering protection against insects while ensuring proper airflow.

By leveraging cost-effective components and intelligent automation, this system not only improves home security and comfort but also reduces manual intervention. The integration of real-time environmental sensing with automated actuation makes this Smart Window System a significant step toward sustainable and intelligent living environments. Compared to traditional window control methods, this innovation provides an efficient, responsive, and user-friendly solution for modern smart homes.

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Chapter 1

Introduction

In today's modern era, intelligent home automation systems are essential for enhancing safety, energy efficiency, and convenience. Traditional ventilation methods require manual operation, which can be inefficient in maintaining indoor air quality and thermal comfort. With advancements in sensor networks and automation technologies, smart home solutions now offer dynamic environmental adaptation without human intervention.

The Smart Window and Mosquito Net Automation System provides an automated solution for home ventilation and rain protection. By integrating sensors such as light-dependent resistors (LDRs), rain sensors, and smoke sensors, the system continuously monitors environmental conditions and transmits real-time data to an Arduino-based control unit. This enables intelligent automation, allowing the window and mosquito net to function dynamically.

A key feature of this system is the smart window mechanism, which opens automatically after detecting smoke or gas, ensuring proper ventilation and reducing suffocation hazards. In addition, the system closes the window when rain is detected, helping to maintain indoor conditions effectively.

This project follows a structured approach that includes a review of the literature, sensor-based data collection, algorithm development, and validation through testing. By combining real-time environmental sensing with automation, the system improves home security and comfort, making it a practical and innovative solution for modern smart homes.

Chapter 2

Literature Review

The increasing demand for intelligent home automation solutions has led to extensive research on sensor-driven systems for ventilation, safety, and energy efficiency. Various studies highlight the role of automated smart windows and mosquito nets in enhancing indoor air quality, preventing insect entry, and optimizing climate control. Research by A. Patel (2019) explores the significance of sensor-based automation in improving indoor air quality and energy efficiency. The study emphasizes the role of Light Dependent Resistors (LDRs), rain sensors, and smoke detectors in automating home ventilation, reducing manual intervention, and enhancing energy savings. A study by Rahul Verma and S. Nair (2021) discusses the use of actuator-controlled smart windows and mosquito nets for improved home security and comfort. Their research demonstrates how automated ventilation systems can contribute to health benefits by reducing air pollutants and maintaining optimal airflow.

Similarly, K. Gupta et al. (2020) investigate smart window systems using IoT-based automation. Their study presents a framework where environmental sensors detect real-time conditions and transmit data to a microcontroller, which dynamically adjusts window positions for optimal airflow and temperature regulation. A study by T. Nakamura et al. (2018) focuses on climate-adaptive home automation, where intelligent systems regulate airflow, detect pollutants, and respond to environmental changes to maintain an energy-efficient indoor atmosphere. The research highlights the importance of integrating multiple sensors to ensure real-time data collection and automated decision-making. Additionally, research by D. Martinez et al. (2022)

examines smart mosquito net systems that deploy based on light intensity and user-defined parameters. Their findings indicate that automated insect prevention significantly reduces health risks associated with mosquitoes and improves user convenience.

Several commercial smart home systems, such as Google Nest, EcoBee, and SmartVent, utilize sensor-driven automation to regulate indoor climate and ventilation. While Google Nest and EcoBee primarily focus on temperature and air quality management, SmartVent offers a comprehensive solution with automated window and vent control. However, these solutions are often costly and require complex installations. Our Smart Window and Mosquito Net Automation System integrates similar sensor-based automation principles while providing a cost-effective and efficient alternative. By leveraging real-time environmental sensing and automated actuation, our system ensures enhanced home safety, energy efficiency, and user convenience. This literature review underscores the growing importance of smart home automation and the potential of sensor-driven ventilation systems in creating safer and more comfortable living environments.

Chapter 3

System Development

This chapter details the development of the Smart Window and Mosquito Net Automation System, including the hardware components, software implementation, and integration of various sensors for automation. The system is designed to dynamically respond to environmental conditions, enhancing home safety, energy efficiency, and comfort. The development process involves selecting appropriate components, designing circuits, writing control algorithms, and testing the system for efficiency. This chapter provides an in-depth discussion on system architecture, functional modules, and implementation strategies.

3.1 Objectives

- To automate window control based on rain, smoke, and AC status.
- To deploy the mosquito net in low light for insect prevention and airflow.
- To enhance safety by ensuring ventilation during smoke or gas leaks.
- To improve energy efficiency by adjusting windows for optimal indoor climate.
- To provide seamless user convenience with sensor-based automation.

3.2 Block Diagram

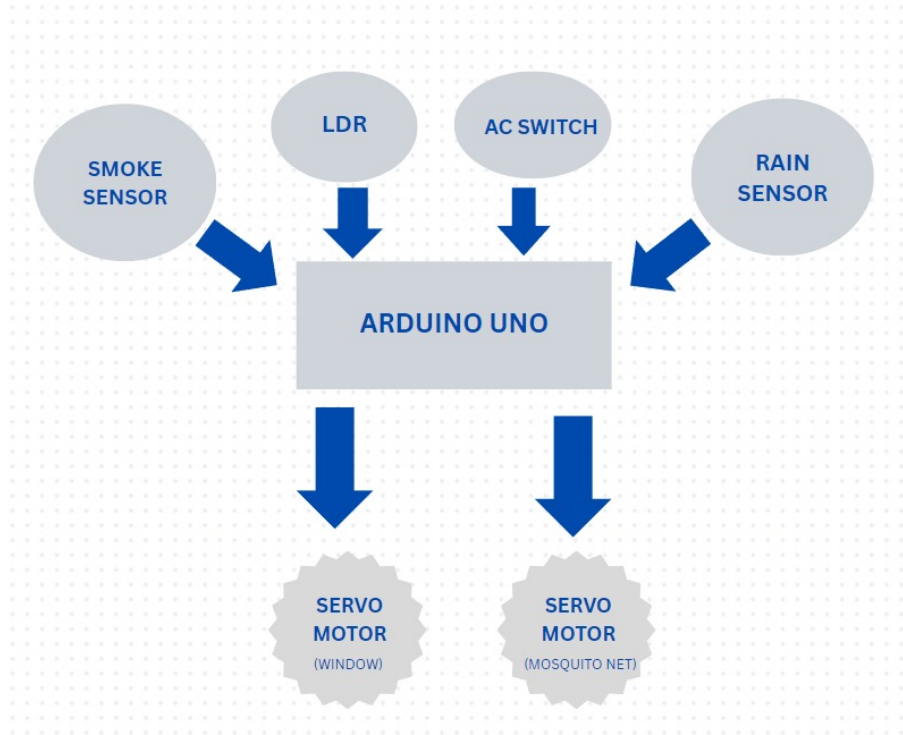


Figure 3.1: Block Diagram Of the Proposed Idea

The block diagram represents the Smart Window Automation System, where Arduino Uno acts as the central controller, processing inputs from multiple sensors to automate window and mosquito net operations. The smoke sensor detects smoke or gas leaks, triggering the window to open for ventilation, while the rain sensor ensures the window closes during rainfall to prevent water entry. The LDR monitors ambient light intensity, retracting the mosquito net during daylight and deploying it at night for insect protection. An AC switch detects whether the air conditioner is on, keeping the window closed for energy efficiency or opening it for natural ventilation when the AC is off. Based on real-time data, the Arduino controls servo motors—one for opening/closing the window and another for deploying/retracting the mosquito net. This automated system enhances home safety, comfort, and energy efficiency, reducing the need for manual intervention.

3.3 Hardware Components

3.3.1 ARDUINO UNO

The Arduino Uno is a widely used microcontroller board based on the ATmega328P chip, operating at 16MHz. It has 14 digital input/output pins, 6 analog inputs, and supports PWM output for motor and sensor control. It is programmable using the Arduino IDE (based on C/C++) and facilitates seamless interfacing with a variety of sensors and actuators.

The Arduino Uno is the central processing unit of the system, responsible for collecting, processing, and executing commands based on sensor data. It receives input from various sensors, analyzes environmental conditions, and sends appropriate control signals to actuators such as the servo motor and relay. The microcontroller processes real-time data to determine whether to open or close the window. Its user-friendly programming environment, versatility, and sufficient I/O pins make it an ideal choice for this automation system.

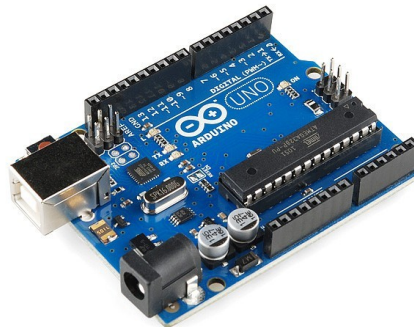


Figure 3.2: Arduino Uno

3.3.2 RAIN SENSOR

The rain sensor detects the presence of rain and sends an electrical signal to the microcontroller. When rain is detected, the Arduino Uno processes this input and triggers the servo motor to close the window, preventing water from entering. The sensor consists of a conductive plate that detects water droplets, creating a signal



Figure 3.3: Rain Sensor

that is interpreted by the microcontroller. Additionally, it can be calibrated to detect varying levels of rainfall intensity, ensuring that the system responds appropriately. This feature helps in preventing unnecessary window closures due to minimal drizzles while ensuring effective protection during heavy rainfall. The rain sensor plays a crucial role in automating climate adaptation by safeguarding indoor spaces from water damage and maintaining optimal indoor conditions.

3.3.3 SMOKE SENSOR

A smoke sensor is used to detect the presence of smoke or harmful gases in the environment, making it an essential component in fire detection systems and air quality monitoring. In this project, the smoke sensor can be integrated with an Arduino Nano to trigger alerts or activate safety mechanisms when smoke levels exceed a threshold. The sensor outputs an analog or digital signal to the microcontroller, which then processes the data and takes necessary actions, such as turning on an alarm, activating ventilation, or sending notifications. A common smoke sensor, such as the MQ-2, operates at 5V DC and consumes approximately 150mA of power. It can detect gases like LPG, carbon monoxide (CO), alcohol, propane, hydrogen, and methane, with a response time of less than 10 seconds. The sensor provides both analog and digital outputs, allowing for flexible interfacing with microcontrollers. Additionally, it features a built-in potentiometer for sensitivity adjustment, ensuring accurate detection



Figure 3.4: Smoke Sensor

based on environmental conditions. The MQ-2 smoke sensor is widely used in IoT-based fire detection systems, smart homes, and industrial safety applications. When smoke levels rise, the sensor detects the change in air composition and sends a signal to the Arduino, enabling automated response actions to enhance safety and prevent hazards.

3.3.4 LDR (LIGHT DEPENDENT RESISTOR)

A Light Dependent Resistor (LDR) is a sensor that detects changes in light intensity. It is commonly used in automatic lighting systems, solar tracking, and security applications. The resistance of an LDR decreases as light intensity increases, allowing it to generate an analog signal that can be read by a microcontroller like the Arduino Nano. In this project, the LDR helps monitor ambient light levels, enabling automation such as switching lights on or off based on the detected brightness. The LDR is typically connected in a voltage divider circuit with a 10k resistor, providing an output voltage that varies with light intensity. It operates on 3.3V to 5V DC and has a fast response time to changes in light conditions. The sensitivity of the sensor depends on the material used, with common types being cadmium sulfide (CdS) photoconductive cells. LDRs are widely used in energy-efficient applications where automatic adjustments are needed based on natural light. In this project, when the LDR detects low light levels, the Arduino can activate a relay to turn on a light or



Figure 3.5: Light Dependent Resistor (LDR)

trigger other responses, making it useful for smart home automation or environmental monitoring systems.

3.3.5 SERVO MOTOR (for Window and Mosquito Net Control)

Servo motors are used for precise window and mosquito net movement based on sensor inputs. One servo motor controls the window, ensuring it opens or closes automatically in response to rain, smoke, and AC status. Another servo motor operates the mosquito net, deploying it at night and retracting it during the day. Servo motors are energy-efficient, accurate, and durable, making them ideal for precise mechanical movements in smart home systems.



Figure 3.6: Servo Motor

3.4 Software Used

3.4.1 ARDUINO IDE

The Arduino Integrated Development Environment (IDE) is a software platform used for programming Arduino microcontrollers. It provides a user-friendly interface for writing, compiling, and uploading code to Arduino boards.

1. **Writing Code:** The Arduino IDE allows you to write code in the Arduino programming language, which is a simplified version of C/C++. It provides a range of built-in functions and libraries for interacting with hardware components, making it easy to get started with programming.
2. **Compiling:** Once you've written your code, you can compile it within the Arduino IDE. Compilation checks for errors and converts your code into machine-readable instructions that can be understood by the Arduino board.
3. **Uploading:** After compiling, you can upload your code to an Arduino board connected to your computer via USB. The IDE handles the process of transferring the compiled code to the board's memory, where it can be executed.
4. **Serial Monitor:** The Arduino IDE includes a Serial Monitor tool that allows you to send and receive data between your Arduino board and your computer over a serial connection. This is useful for debugging and monitoring the behavior of your code.

3.5 Methodology and Implementation

The proposed system is designed to automate window and mosquito net control using real-time sensor data. The methodology involves the integration of sensors, actuators, and a microcontroller to achieve smart home automation with minimal user intervention.

3.5.1 System Operation

- Sensors continuously monitor environmental conditions, including light intensity, rain detection, smoke presence, and air conditioning status.
- The Arduino Nano processes sensor data and applies predefined decision-making logic.

- Based on sensor inputs, the system executes the following actions:
 - If high ambient light is detected, the mosquito net retracts; if low light is detected, it deploys.
 - If rainfall is detected, the window automatically closes to prevent water entry.
 - If smoke or gas is detected, the window opens immediately for ventilation.
 - If the AC is turned on, the window remains closed; if the AC is off, the window opens for natural ventilation.
- A manual override switch allows users to bypass automation and control the window and mosquito net manually.
- All decisions are executed in real-time, ensuring efficient system operation and immediate responses to changing environmental conditions.

3.5.2 Implementation Process

- The system is programmed using the Arduino IDE in C/C++.
- Sensor data is continuously read and processed by the Arduino Nano.
- The decision-making algorithm determines the appropriate actuator response.
- Servo motors are controlled using PWM signals to adjust the window and mosquito net.
- The system is tested under different environmental conditions to ensure proper functionality.

This approach ensures an efficient and automated solution for home ventilation and mosquito net control while maintaining flexibility through manual user control.

3.6 Circuit Diagram

The circuit diagram represents an automated window and mosquito net control system using an Arduino Uno, sensors, servo motors, and manual switches. The Arduino Uno

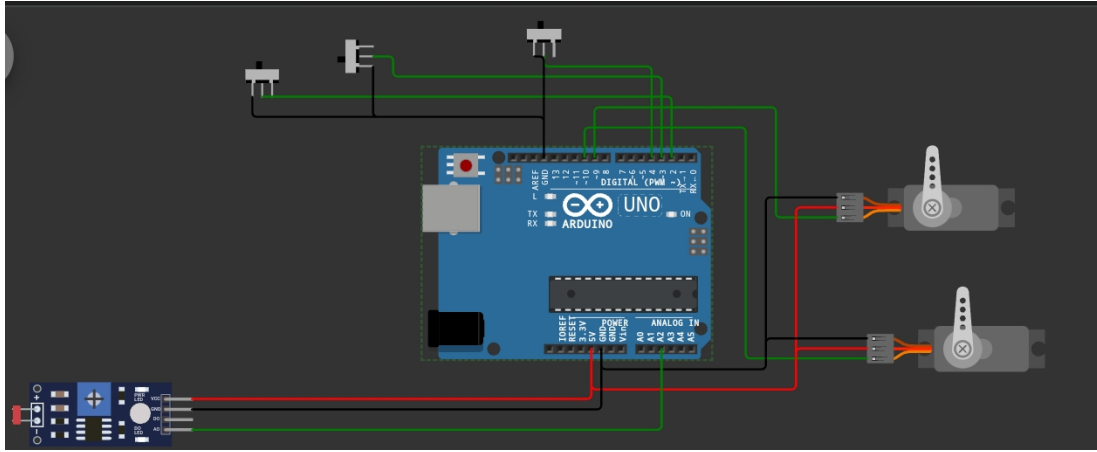


Figure 3.7: Circuit Diagram of The Proposed Idea

acts as the central processing unit, receiving real-time sensor inputs and controlling the servo motors accordingly. The system includes multiple sensors such as an LDR, which detects ambient light to control the mosquito net movement, and a rain sensor, which detects rainfall and signals the Arduino to close the window to prevent water entry. Additionally, there are two toggle switches that function as manual overrides, allowing users to open or close the window and mosquito net manually.

The circuit incorporates two servo motors responsible for moving the window and mosquito net. Each servo motor has three connections: an orange wire for the signal (connected to the Arduino's PWM digital pins), a red wire for power (connected to the 5V supply of the Arduino), and a black wire for ground (GND). The entire system is powered using the Arduino's 5V and GND pins, ensuring proper voltage distribution to all components.

The working mechanism involves the rain sensor detecting water and signaling the Arduino to activate the servo motor, closing the window automatically. Similarly, the LDR sensor detects changes in light intensity and controls the mosquito net accordingly. If the room is dark, the mosquito net is deployed for protection, whereas in daylight, it is retracted. The manual override switches provide flexibility, allowing users to bypass automation and control the window and mosquito net as needed. This intelligent system enhances home automation by improving air quality, ensuring comfort, and reducing manual intervention.

Chapter 4

Conclusion

The Smart Window and Mosquito Net Automation System successfully integrates sensor-based automation to enhance indoor air quality, energy efficiency, and user convenience. By utilizing real-time environmental data from LDR, rain, smoke, and AC status sensors, the system effectively controls window and mosquito net operations without requiring manual intervention. The ability to automatically open windows for ventilation in response to smoke or gas leaks, close windows during rainfall, and adjust the mosquito net based on light conditions ensures a safer and more comfortable living environment. Additionally, the incorporation of an AC status switch helps in optimizing energy consumption by maintaining indoor temperature efficiently.

The system was tested under various environmental conditions, and the results demonstrated its reliability and responsiveness. The automation provided seamless functionality while also allowing users to override the system manually when necessary. However, certain limitations such as the lack of remote access and dependence on sensor accuracy highlight areas for future improvement. Enhancing the system with IoT connectivity for remote monitoring and AI-based decision-making could further improve its adaptability and performance.

Overall, this project presents a practical and innovative approach to smart home automation, contributing to improved air circulation, energy conservation, and user safety. The Smart Window and Mosquito Net Automation System offers a scalable solution for modern homes and buildings, aligning with the growing need for intelligent, energy-efficient, and automated living spaces.

Chapter 5

Future Scope

The future scope of this project includes integrating IoT for remote monitoring and control via a mobile app, enabling real-time alerts and user accessibility. Implementing AI and machine learning can optimize automation by analyzing environmental conditions and user preferences, improving efficiency. Enhancing sensor accuracy will ensure reliable performance and reduce false triggers.

Expanding the system for larger buildings and industrial applications can improve air quality management on a broader scale. Voice control integration with smart home assistants like Google Assistant or Alexa can further enhance convenience. Solar-powered operation can make the system more energy-efficient and sustainable.

Future improvements could also include smart glass technology for automatic transparency adjustment, reducing energy consumption. Adding dust and pollution sensors would enhance air quality monitoring. With these advancements, the system can become a more comprehensive, energy-efficient, and adaptable smart home solution.

References

1. Stolojescu-Crisan, Cristina, Calin Crisan, and Bogdan-Petru Butunoi. "An IoT-based smart home automation system." *Sensors* 21.11 (2021): 3784.
2. Vijayan, D. S., et al. "Automation systems in smart buildings: a review." *Journal of Ambient Intelligence and Humanized Computing* (2020): 1-13.
3. Kodali, Ravi Kishore, et al. "IoT based smart security and home automation system." 2016 international conference on computing, communication and automation (ICCCA). IEEE, 2016.
4. Gunge, Vaishnavi S., and Pratibha S. Yalagi. "Smart home automation: a literature review." *International Journal of Computer Applications* 975.8887-8891 (2016).
5. Vishwakarma, Satyendra K., et al. "Smart energy efficient home automation system using IoT." 2019 4th international conference on internet of things: Smart innovation and usages (IoT-SIU). Ieee, 2019.
6. Jabbar, Waheb A., et al. "Design and implementation of IoT-based automation system for smart home." 2018 International symposium on networks, computers and communications (ISNCC). IEEE, 2018.
7. Chasta, Rohit, et al. "A smart building automation system." *International Journal of Smart Home* 10.8 (2016): 91-98.
8. Singh, Himanshu, et al. "IoT based smart home automation system using sensor node." 2018 4th International Conference on Recent Advances in Information Technology (RAIT). IEEE, 2018.